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PalmSource Seminar Series
Build Tomorrow ▸▸▸ Today

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WebLink Wireless

palm™ POWERED

WebLink Wireless

Powered by WebLink WIRELESS™

Introducing the H-500 Wireless Palm

With wireless …

THERE IS NO SILVER BULLET!

There are always compromises

Variables

- **Quantity of data**
- **Latency**
- **Coverage**
- **In-building coverage**
- **Battery life**
- **Cost**
  - Device
  - Airtime
- **Reliability**
- **Form Factor**

Successful wireless applications match use with variables.

- **Established July of 1989**
- **Headquartered in Dallas, TX**
- **1.0 Million units in service**
- **Privately-owned**
  - Leucadia National Corp.
- **Operates the nation's largest wireless data network**
  - ReFLEX Narrowband PCS



Introducing the Hunetec H-500



- **Combining Palm OS 5 with embedded ReFLEX radio**
- **Operates on WebLink Wireless and other ReFLEX networks**

Hunetec H-500 Wireless PDA



- **QWERTY keyboard**
- **High resolution color display**
- **16MB SDRAM**
- **Rechargeable battery**
- **SD/MMC card slot**
- **Processor: Motorola DragonBall MX-1 (150MHz)**
- **Touch screen with stylus**
- **Wireless email and messaging on ReFLEX protocol**

Hunetec H-500 – Target Segments



- **Field service organizations**
- **IT departments**
- **Mobile professionals**

- **In the following verticals:**
  - Manufacturing
  - Healthcare
  - Utilities
  - Information
  - Finance and Insurance
  - Professional Services
  - Hospitality
  - Government

Hunetec H-500 – Developer Opportunities



- **Wireless extension of existing application**
- **New, wireless applications to existing customer set**

Possible Application Types

- **Wireless forms**
  - Dispatch operations
  - Trouble ticket management
- **Data query from device**
  - Inventory levels
  - Job status
  - Other database queries
- **Updating databases**
  - Sales orders
  - Time cards

Wireless Advantages of ReFLEX

- **“Always on” for constant connectivity**
- **Battery life of 5 to 7 days, depending on usage**
- **Superior in-building penetration**
- **Simulcasting from multiple sites at once**
  - Unlike cellular, broadband PCS or Wi-Fi
- **“Store and Forward” technology**
- **Broad geographic/international coverage**

Wireless Advantages of ReFLEX

Delivering the highest reliability for mission-critical business applications

on a mature, stable technology platform



The Coverage Advantage

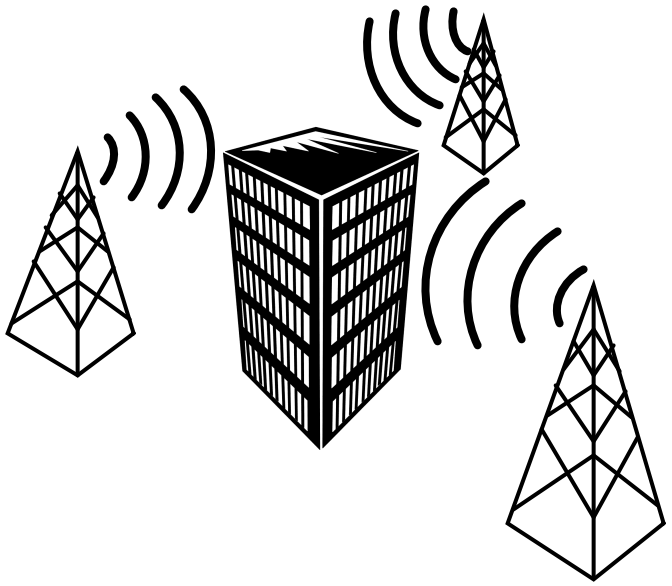
- **Wireless data network covering over 90% of U.S. population**
- **Coverage in Canada and Mexico**
- **Red - 2way / 1way**

  Blue - 1way International

Superior In-Building Penetration

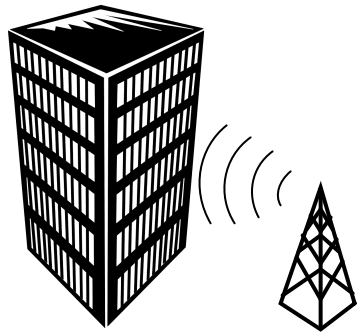
ReFLEX networks:

- Simulcast from multiple towers
- Transmitters high off ground (up to 300 ft)
- High power (1000 watts ERP*)



Cellular-type networks (including Blackberry):

- Transmit from a single tower
- Low to the ground (100 ft)
- Weak power (60 watts ERP*)
- Effects both voice and data signals



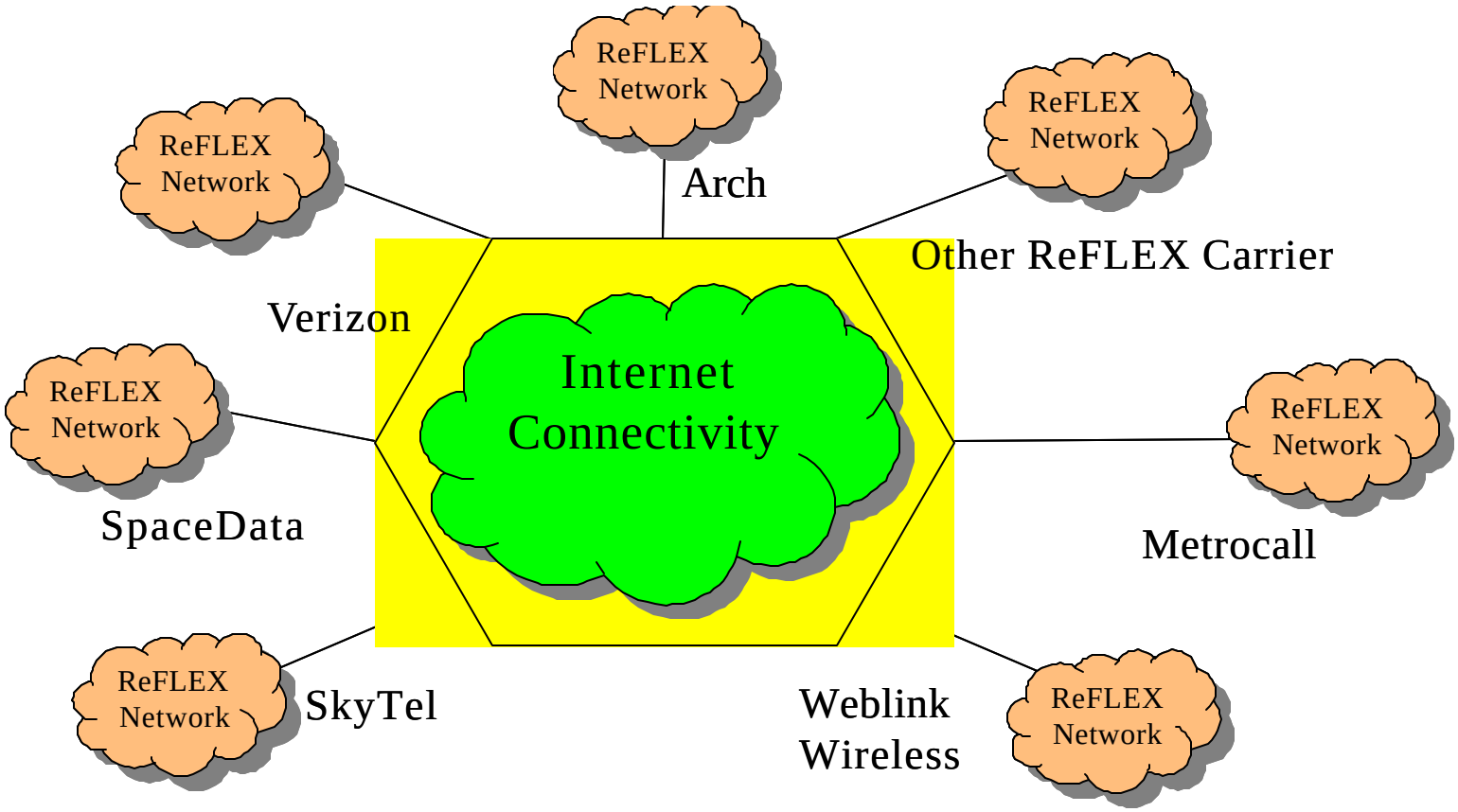
* ERP = Effective Radiating Power

Rules of ReFLEX

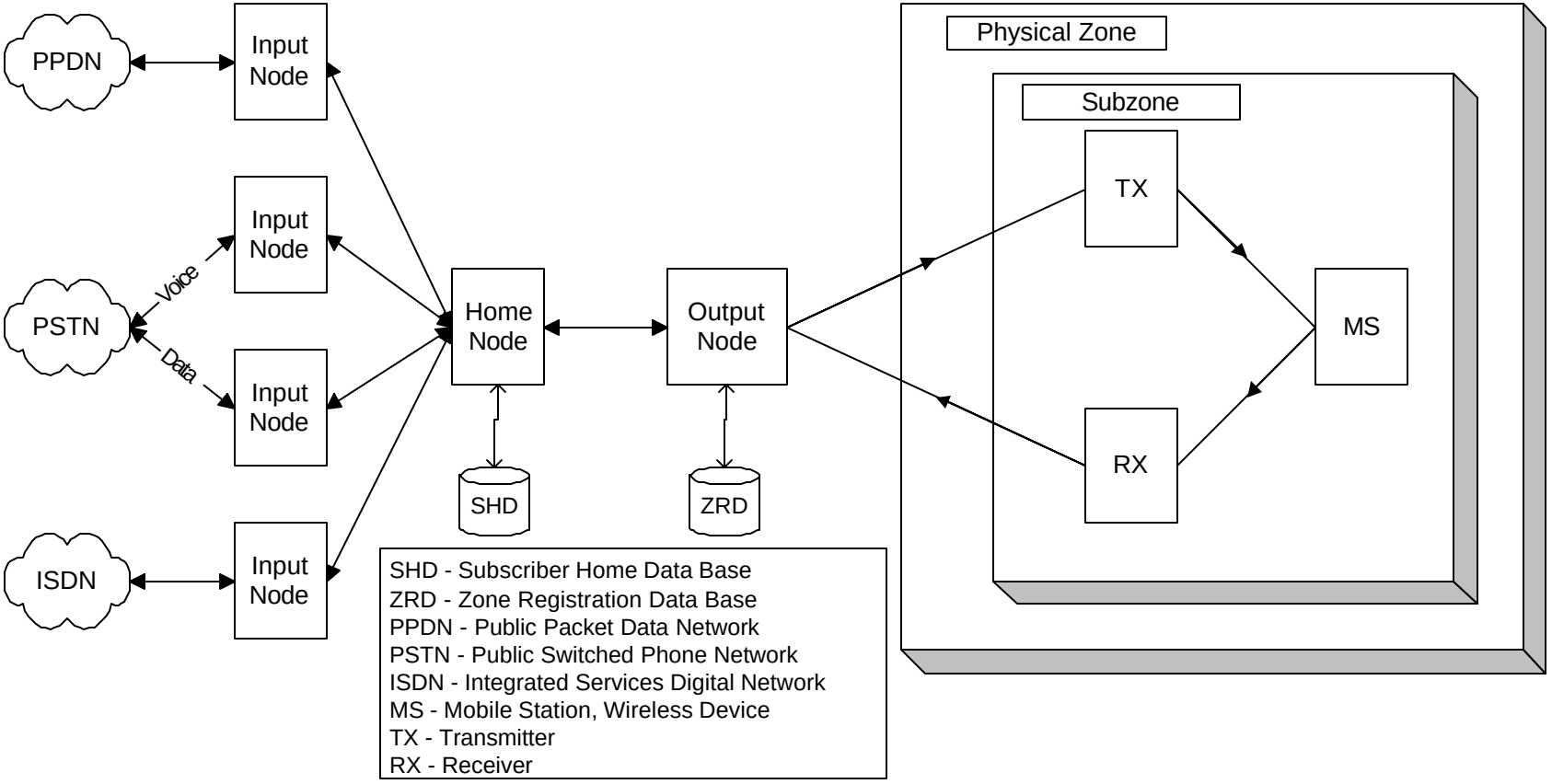
- **Message size < 1Kbyte per transmission**
  - Larger messages can be fragmented and reassembled
- **Average network throughput per subscriber of 100 Kbytes per month**
- **Latency of 1 minute each way**
- **Connectivity via WCTP or SMTP**
- **Asynchronous transmission**
- **Latency of ~ 1 minute roundtrip**

WCTP Protocol

Open Architecture via Internet Standards



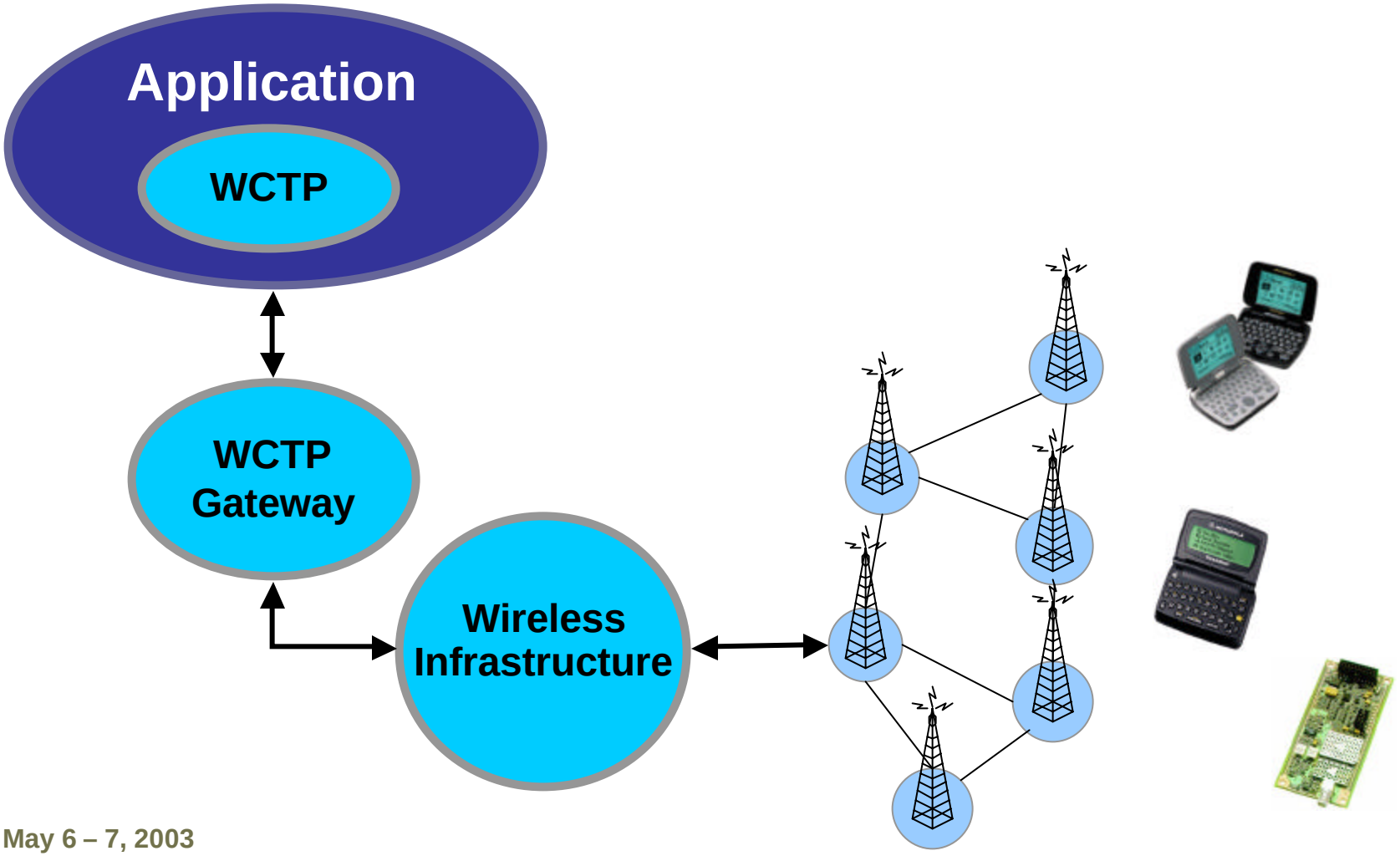
Elements of a NPCS/ReFLEX Network



What is WCTP?

- **The *preferred* standard protocol for sending & receiving messages from the Internet**
- **Designed for wide range of messaging and wireless devices, not only ReFLEX™**
- **Provides an extensible architecture**
- **Supports transport of any type of binary data**

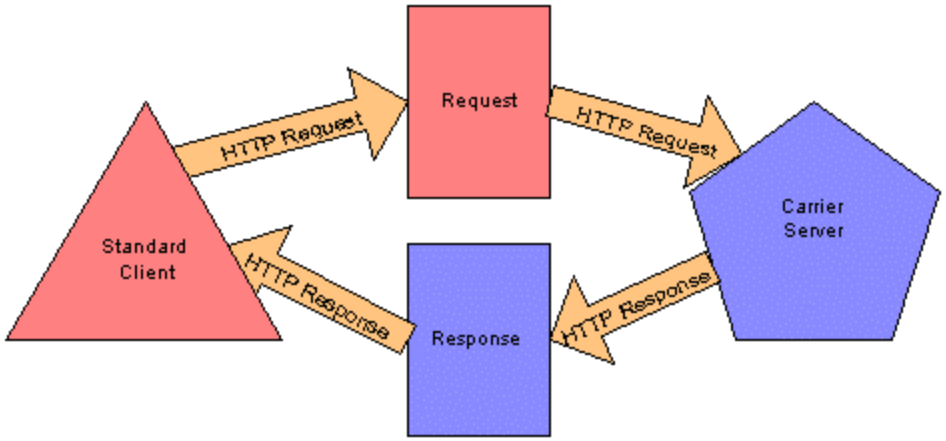
Ok, How's it work?



Tell me more?

- **Supports bi-directional communications**
- **Based on XML and HTTP transported**
- **It is the application interface to the wireless network**
- **Five actors**
  - Carrier Gateway (named client or server)
  - Enterprise Host (named client or server)
  - Polling Enterprise (named client only)
  - Proxy (named client or server)
  - Transient Client (typically unnamed client only)

Transport Model

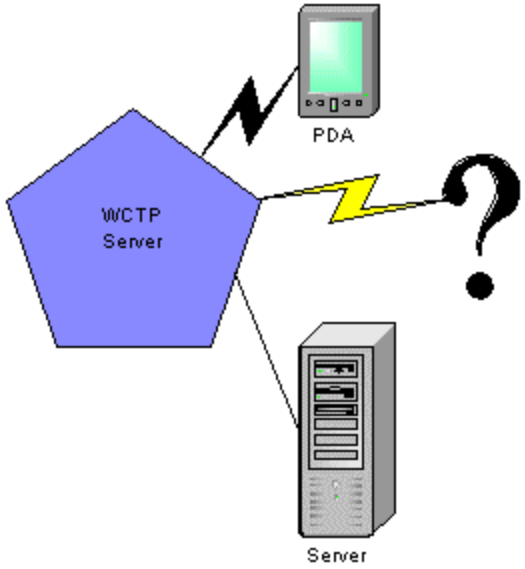


HTTP PORT 80

Looks simple enough.

- **Hides the complexity of the wireless network infrastructure, protocols, and devices**
- **Maintained by Paging Technical Committee**
  - Affiliated with AAPC
  - Available at http://www.wctp.org
  - Working group at Yahoo groups: PTC-PWG-WCTP

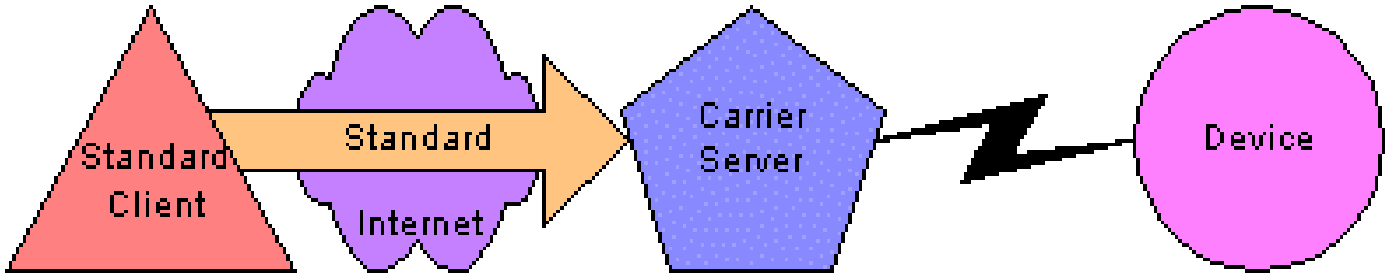
Device Flexibility



Pagers, Handsets, Embedded devices

What're the Operations?

Submit Message



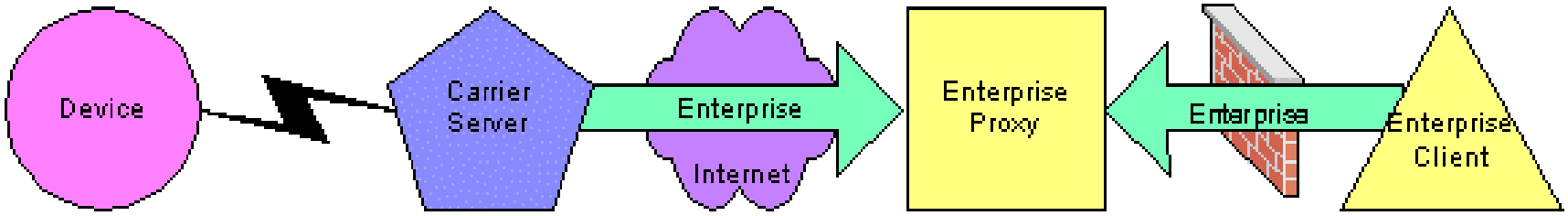
Anonymous user sends message to a device

I understand, what others?

- **Get Status**
- **Reply Message**
- **Get Confirmation**
- **Lookup Subscriber**
- **Lookup Response**
- **Get Version**
- **and more**

What about Enterprises & Security?

Pseudo-push behavior



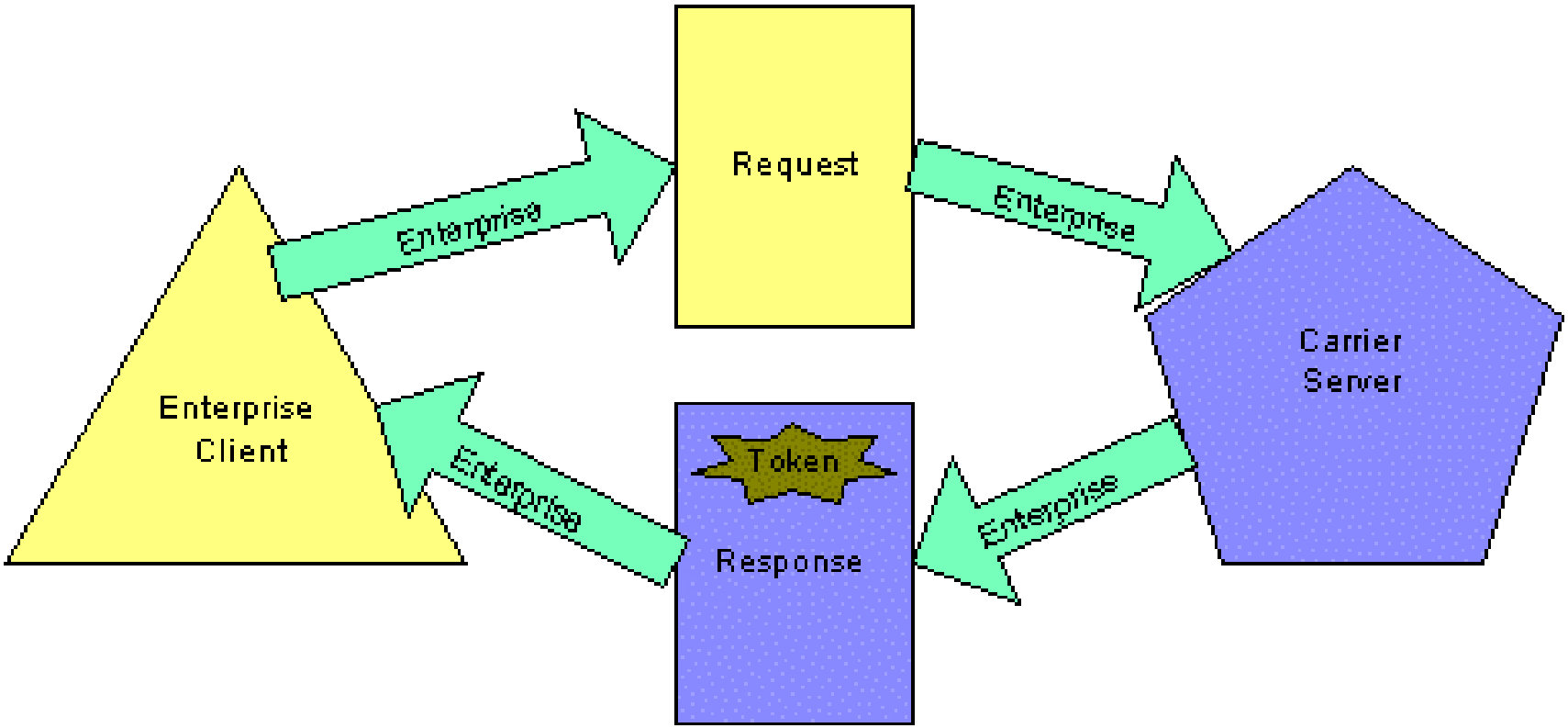
Payloads?

- **Alphanumeric**
- **Opaque**
  - Custom Binary Format
  - Encryption (any scheme)
    - PTC standards for AES (128-256bit keys, ECC PKI)
- **Size Independent**

Messaging Attributes?

- **Preformatted**
- **Truncation**
- **Status Notifications**
- **Response Routes**
- **Multiple Choice Responses**

Version Control

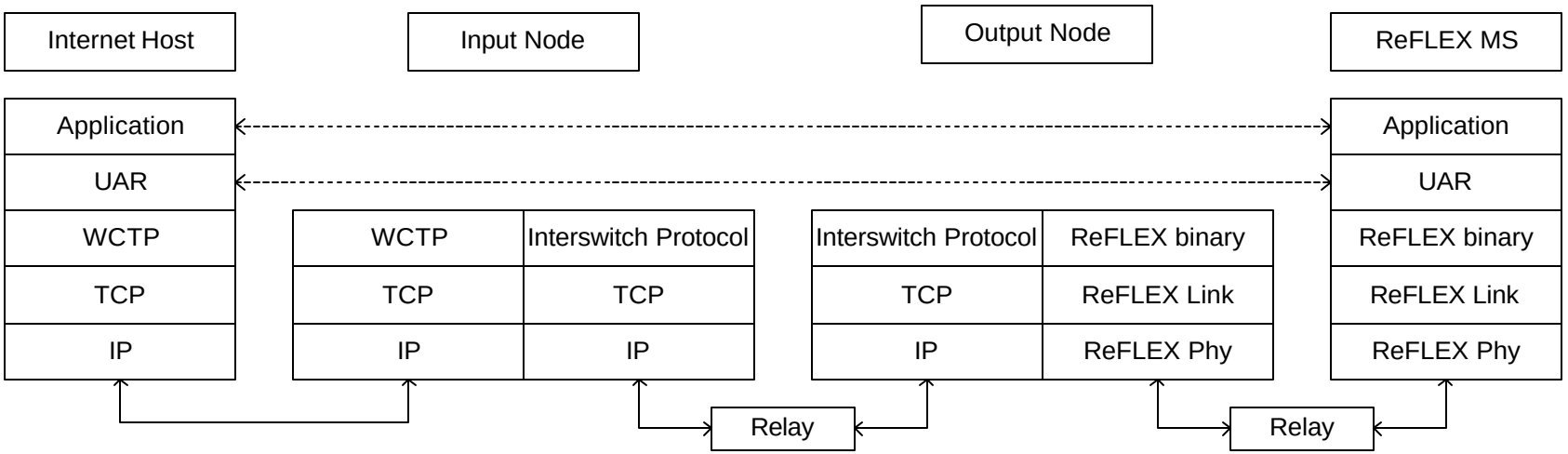


Future Plans for WCTP

- **Enhanced message control attributes**
- **Asynchronous transport support**
- **XML Schemas**
- **Extension framework (location, billing, provisioning, message store)**
- **Current version WCTP 1.2**

Show me a WCTP/UAR Stack model:

Routing payload to/from device



Addressing a message?

[entity]@Transport_address

Transport_address is

- [Protocol:[//]]Domain[:Port][/Path]

For example:

- inventory@myhome.com
- myhome.com/process/inventory.pl
- joe.invent@wctp.myhome.com/
- all go to “wctp.myhome.com”

Addressing a message?

- **Default protocol is http, default host name is “wctp”, default port is 80, default path is “/wctp”.**
- **Could be placed in address book as pager address!**
- **Could be coded into application.**
- **Could be user-configured in app in the field.**

Addressing a message?

- **Entity identifier is**
  - [Scheme:]entity[:Port][/Path]
- **Device “entity” examples:**
  - 8009709325@airmessage.net
  - 8009709325.78295@airmessage.net
  - wctp:john.doe@airmessage.net
  - john_doe/chessapp@airmessage.net

Addressing a message?

- **Host “entity” examples:**
  - inventory@myhome.com
  - joe_user@https:myhome.com/wctp-gw
  - wctp:accounting/receiveables @http://myhome.com:8080/process-wctp
  - im:johndoe@im-gateway.com
- **Any of these are candidates for address book entries!**

UAR? FLEXsuite?

What are these for?

What is FLEXsuite?

- **A collection of I/O, application, & application-enabling protocols**
- **A unique, *recursive* stack model**
  - protocols may carry message, or
  - another FLEXsuite protocol
- **Always used within ReFLEX networks**
  - *can be combined with WCTP for powerful, peer-to-peer transport at application layer*

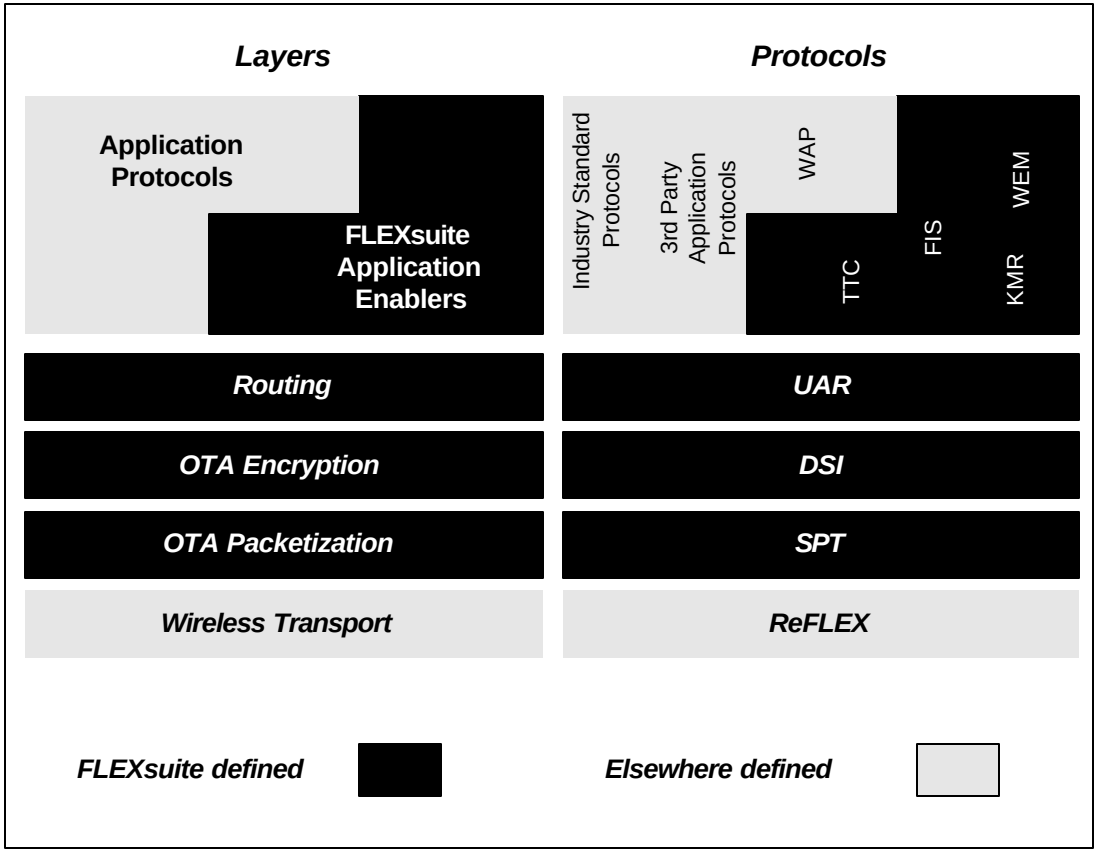
What are the FLEXsuite protocols?

- **Uniform Addressing & Routing (UAR)**
  - mapping PSTN/Internet/ReFLEX networks onto ReFLEX networks
- **Application Data Security Identifier (ADSI)**
  - RC4, AES, ECC encryption
- **Key Management Request Protocol (KMR)**
  - key agreement via ECC Diffie-Hellman

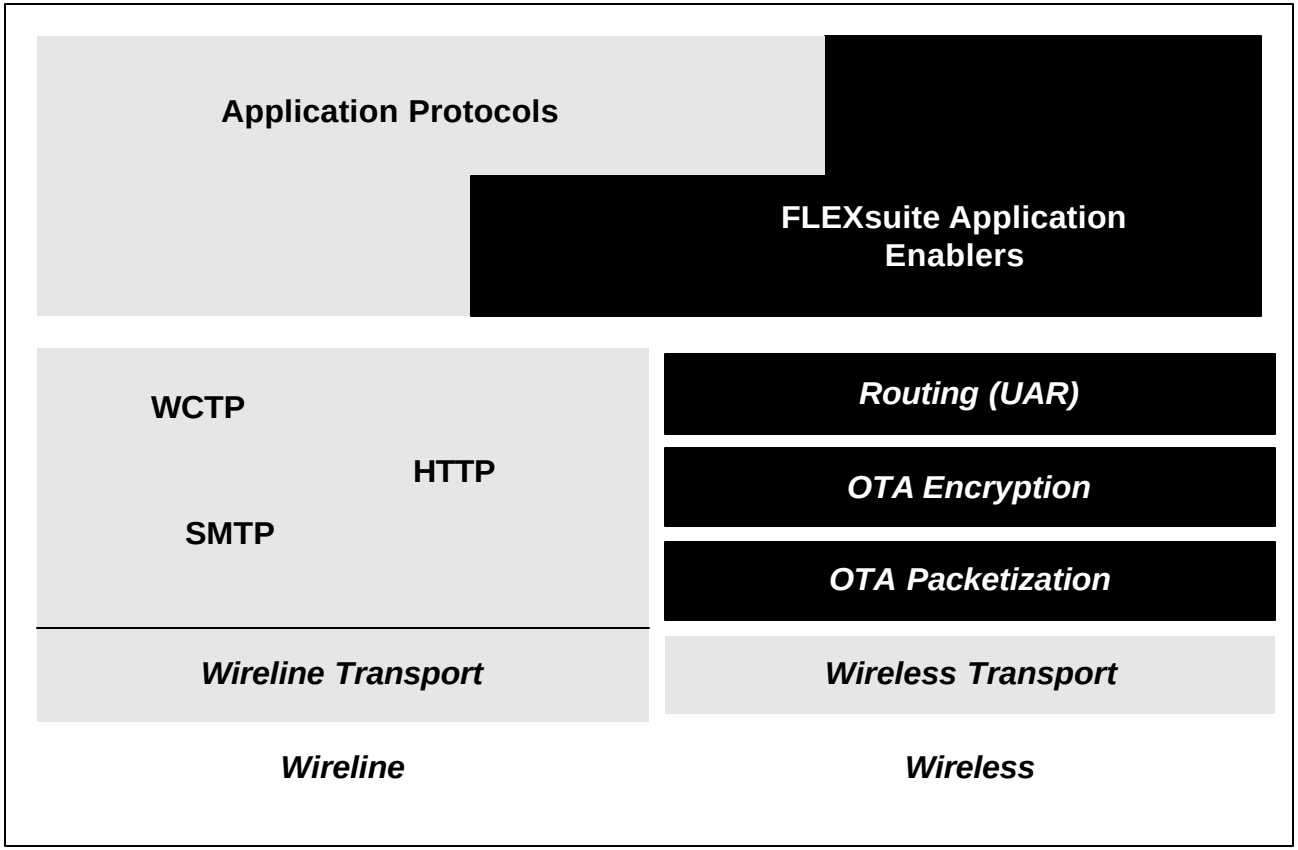
More FLEXsuite protocols

- **Simple Packet Transport (SPT)**
  - message assembly/disassembly
- **Wireless Email (WEM)**
  - RFC-822 ‘like’
  - for transport within UAR
    - just as 822 is encapsulated by RFC-821
- **FLEXInfo Information Service (FIS)**

What about the FLEXsuite Protocol Stack?



What is Content Routing?

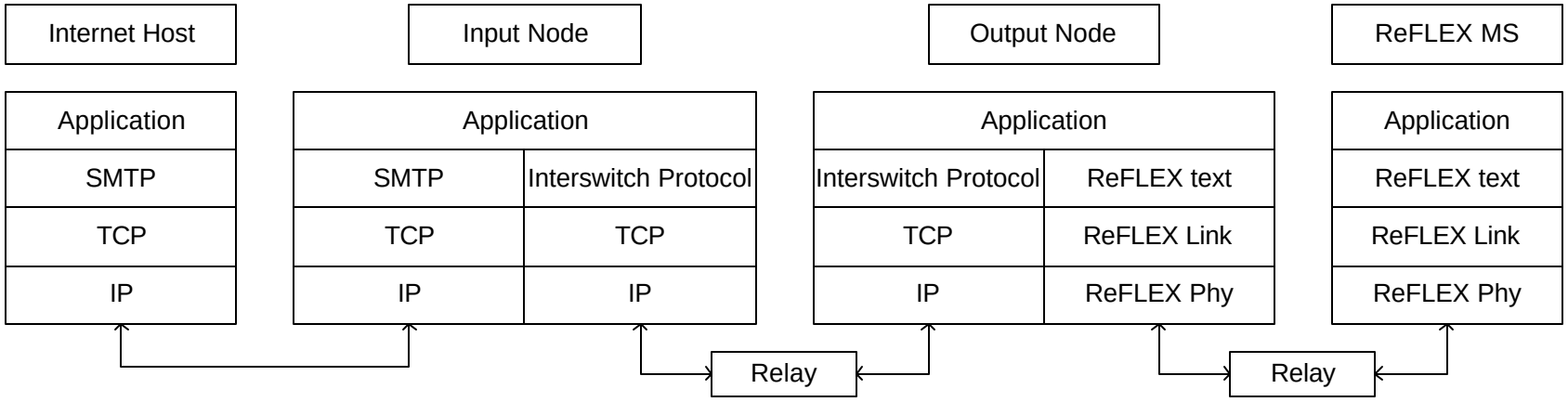


What does UAR do for me?

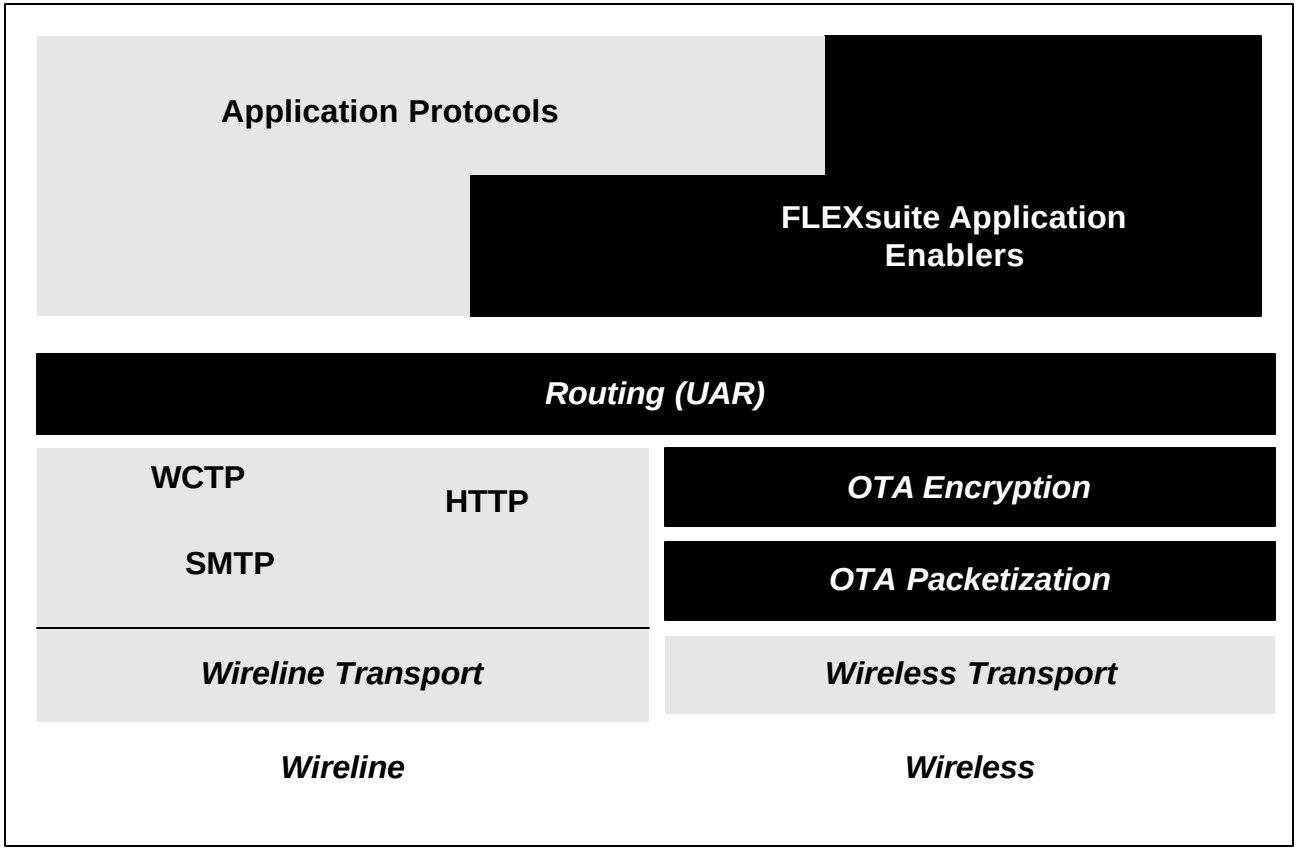
- **Supports, through routing & address adaptation:**
  - ReFLEX Peer-to-peer messaging
  - Wireline device to MS messaging
  - Wireline application server (eg, email) to MS
- **For content routing, activated at an Input Node**
- **For end to end routing, supported at Host**

Content routing with UAR

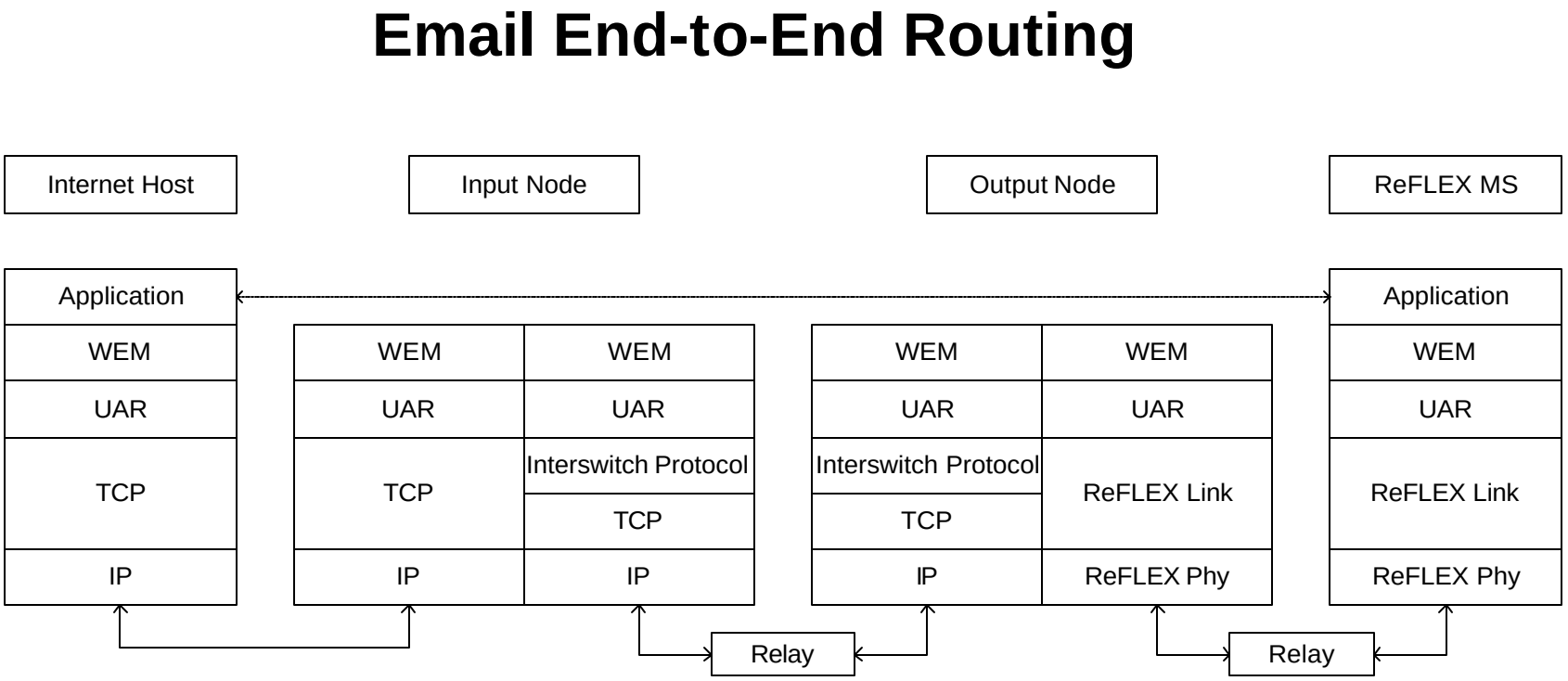
Email Content Routing (SMTP)



What about End-to-End Routing?



UAR for End-to-End routing



How does this get my data to my app?

From Server:

- Submits payload to Carrier via WCTP
  - Recipient ID: user@carrier
  - Sender ID: domain.com/cgi-bin/postMessage.pl
  - Payload: Alphanumeric/Opaque-Binary

From Device:

- Sends payload to Server via WCTP
  - Recipient ID: domain.com/cgi-bin/postMessage.pl
  - Sender ID: user@carrier
  - Payload: Alphanumeric/Opaque-Binary

For Example

From Server:

- Routing on device using UAR:
  - To:
    - Port: 29
    - Path: games/chess/
  - From:
    - Port: 80
    - Path: chess/

Address-types UAR carries?

- **Pageto, for peer to peer, email, wctp**
- **Mailto, for smtp only**
- **Phoneto, for text-to-speech**
- **Faxto, for text-to-document delivery**

Elements of the H-500 SDK

Three functional APIs:

- Application Notification Management
- ReFLEX I/F Management
- Communication Management

Available at Yahoo Groups

- Contact WebLink for information

Aim

- To hide FLEXsuite from the developer

Application Registration Management

AppNotifyRegister

AppNotifyUnregister

GetLibVersion

GetLibAppNotifyList

ReFLEX Module API

ModPower

ModRF

ModTX

ModAlarmMode

ModPrivateMode

ModGetPINNumber

ModGetState

Communication API

CommSend

- RefNum, *data, size, *address, ETE, type

CommReceive

- RefNum, *data, size, type

CommReply

- RefNum, *data, size, ETE, type

CommReturn

- *data, size, type

Next Steps

How to Work With Us

WebLink offers flexibility in distribution and business models

- **Wholesale**
- **Direct**
- **Agent models**

WebLink Wireless Contact Information

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